

Savitzky-Golay optimizer diagnostics

Objective: weighted empirical uncentered second moment of reference-scaled residual body wrench;
no residual mean subtraction

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by
default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly
overridden'}

Selected strict-ZOH solve

status=0 success=False nfev=80 njev=16

message=The maximum number of function evaluations is exceeded.

reported optimality=3.337655968493891e-06

initial={'cost': 842.7235388823646, 'gradient_inf_norm': 0.3992528868852787, 'gradient_l2_norm':
0.4116537044626169, 'jacobian': {'condition_number_nonzero_subspace': 93.33367928077757,
'numerical_rank': 13, 'numerical_tolerance': 3.874280675564741e-10, 'singular_values':
[407.6684347383664, 380.1432051432364, 295.14671076675495, 45.1924442978017,
21.37779972092004, 16.22039530919293, 14.103879147185037, 12.166339300771662,
12.020792104651965, 9.46265573015221, 8.356918421260398, 7.44614946253298,
4.367859896661411, 9.93430660519622e-15]}, 'residual_l2_norm': 41.05419683497327}

final={'cost': 842.7235262339176, 'gradient_inf_norm': 3.337655968493891e-06,
'gradient_l2_norm': 4.484176143224626e-06, 'jacobian': {'condition_number_nonzero_subspace':
93.3238309291668, 'numerical_rank': 13, 'numerical_tolerance': 3.874435583861176e-10,
'singular_values': [407.68473485392946, 380.1585808344055, 295.1587905274326,
45.19430596705722, 21.373209876703203, 16.22133777424083, 14.10509596749835,
12.16761304366537, 12.027654797647761, 9.466137698907506, 8.357751565366835,
7.446636554782577, 4.368495493539736, 1.1568073821019579e-14]}, 'residual_l2_norm':
41.05419652688182}

coordinate_step_l2_norm=0.00047593948895730806

cost_reduction=1.2648446954699466e-05

gradient_tolerance_satisfied=False

finite_difference_max_relative_error=1.1675691017274014e-07

finite_difference_directional_checks=[{'analytic_l2_norm': 20.516676967475735,
'difference_l2_norm': 2.3954639750210094e-06, 'direction': 'negative_gradient',
'finite_difference_l2_norm': 20.516678383120585, 'relative_error': 1.1675691017274014e-07,
'scheme': 'central', 'step': 0.01396464855728762}]

invalid_trial_evaluations={'invalid_evaluation_count': 0, 'invalid_examples': [],
'invalid_exception_counts': {}, 'penalty_residual_l2_norm': 41054196.83497327, 'policy':
'invalid floating-point trial points are assigned a large fixed residual so the trust-region
trial is rejected; the scientific parameter chart itself remains unbounded'}

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.